WORKED SOLUTION

STEP III 2017, Question 5

Pure • Polar Curves

STEP III 2017, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s3-q5

The key idea

Write the curve parametrically, with θ as the parameter: $x = f(\theta)\cos\theta$, $y = f(\theta)\sin\theta$. The gradient then comes from the chain rule, and "meeting at right angles" becomes "the product of the gradients is -1". In the last part the family of curves $r = a(1+\sin\theta)$ passes through every point of C, so the right-angle condition holds all along C. That turns it into a first-order differential equation for f.

Solution

The gradient

On the curve, $x = f(\theta)\cos\theta$ and $y = f(\theta)\sin\theta$. Differentiating with respect to θ by the product rule,

$$\frac{\mathrm{d}x}{\mathrm{d}\theta} = f'(\theta)\cos\theta - f(\theta)\sin\theta, \qquad \frac{\mathrm{d}y}{\mathrm{d}\theta} = f'(\theta)\sin\theta + f(\theta)\cos\theta.$$

By the chain rule, and then dividing top and bottom by $\cos\theta$ (assuming $\cos\theta \neq 0$),

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{f'(\theta)\sin\theta + f(\theta)\cos\theta}{f'(\theta)\cos\theta - f(\theta)\sin\theta} = \frac{f'(\theta)\tan\theta + f(\theta)}{f'(\theta) - f(\theta)\tan\theta}.$$

Curves meeting at right angles

At a meeting point both curves are at the same θ. Write $t = \tan\theta$, and write f, f', g, g' for the values there. If neither tangent is vertical, the curves are perpendicular exactly when the product of the gradients is -1:

$$(f't + f)(g't + g) = -(f' - ft)(g' - gt).$$

Expanding both sides, the terms in $f'gt$ and $fg't$ cancel, which leaves

$$f'g't^2 + fg + f'g' + fgt^2 = 0 \iff (1+t^2)(f'g' + fg) = 0.$$

Since $1 + t^2 = \sec^2\theta > 0$, this gives $f'(\theta)g'(\theta) + f(\theta)g(\theta) = 0$, as required.

> **Ask yourself** What if one tangent is vertical, or $\cos\theta = 0$, so that a gradient or $\tan\theta$ doesn't exist?

Use direction vectors instead of gradients. The tangent to $r = f(\theta)$ has direction $(f'\cos\theta - f\sin\theta,\ f'\sin\theta + f\cos\theta)$, and similarly for g. Their scalar product is

$$(f'\cos\theta - f\sin\theta)(g'\cos\theta - g\sin\theta) + (f'\sin\theta + f\cos\theta)(g'\sin\theta + g\cos\theta) = f'g' + fg,$$

because the cross terms cancel and $\cos^2\theta + \sin^2\theta = 1$. So perpendicularity is the same condition in every case.

Finding $f(\theta)$

For $r = a(1 + \sin\theta)$ we have $g'(\theta) = a\cos\theta$.

> **Ask yourself** The condition only holds where the curves meet. Why can we use it at every point of C?

Take any point of C, at angle θ with $f(\theta) > 0$ and $\sin\theta \neq -1$. The choice $a = f(\theta)/(1 + \sin\theta) > 0$ gives a member of the family that passes through that point at the same θ. By hypothesis this curve meets C at right angles, so

$$f'(\theta)\, a\cos\theta + f(\theta)\, a(1 + \sin\theta) = 0.$$

Dividing by $a > 0$, every such θ satisfies

$$f'(\theta)\cos\theta + f(\theta)(1 + \sin\theta) = 0.$$

Multiply by $1 - \sin\theta$ and use $1 - \sin^2\theta = \cos^2\theta$:

$$\cos\theta\,[f'(\theta)(1 - \sin\theta) + f(\theta)\cos\theta] = 0.$$

Where $\cos\theta \neq 0$, the bracket vanishes. By the quotient rule the bracket is $(1 - \sin\theta)^2$ times the derivative of $f(\theta)/(1 - \sin\theta)$:

$$\frac{\mathrm{d}}{\mathrm{d}\theta}\left(\frac{f(\theta)}{1 - \sin\theta}\right) = \frac{f'(\theta)(1 - \sin\theta) + f(\theta)\cos\theta}{(1 - \sin\theta)^2} = 0.$$

So $f(\theta) = k(1 - \sin\theta)$ for a constant k, on the interval $-\frac{1}{2}\pi < \theta < \frac{1}{2}\pi$. (Equivalently, we could separate variables in $f'/f = -(1 + \sin\theta)/\cos\theta = -\cos\theta/(1 - \sin\theta)$.) The curve passes through $r = 4$, $\theta = -\frac{1}{2}\pi$, so by continuity $k(1 + 1) = 4$, giving $k = 2$. On $\frac{1}{2}\pi < \theta < \frac{3}{2}\pi$ the same argument applies, and $\theta = \frac{3}{2}\pi$ is the same direction as $-\frac{1}{2}\pi$, so again $k = 2$. Hence

$$\boxed{f(\theta) = 2(1 - \sin\theta)}$$

This is another cardioid, the reflection in the x-axis of $r = 2(1 + \sin\theta)$.

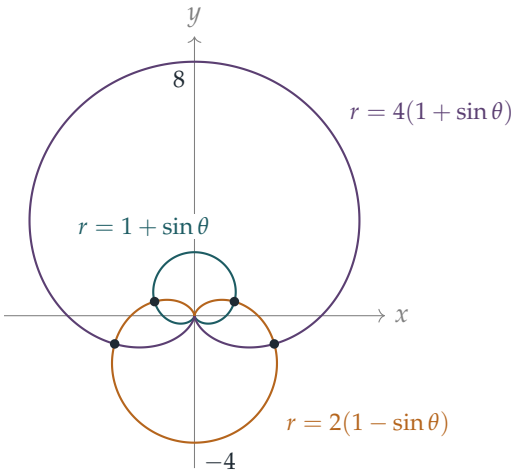
The sketch

The curve $r = a(1+\sin\theta)$ is a cardioid with its cusp at the origin, symmetric about the y-axis. It reaches up to $(0, 2a)$ and crosses the x-axis at $(\pm a, 0)$. The curve $r = 2(1-\sin\theta)$ is a cardioid of the same shape opening downwards: its cusp is at the origin, it crosses the x-axis at $(\pm 2, 0)$ and it reaches down to $(0, -4)$.

The meeting points come from solving $a(1+\sin\theta) = 2(1-\sin\theta)$, which gives $\sin\theta = \frac{2-a}{2+a}$:

- $a = 1$: $\sin\theta = \frac{1}{3}$, $r = \frac{4}{3}$, at the points $\left(\pm\frac{8\sqrt{2}}{9}, \frac{4}{9}\right)$;
- $a = 4$: $\sin\theta = -\frac{1}{3}$, $r = \frac{8}{3}$, at the points $\left(\pm\frac{16\sqrt{2}}{9}, -\frac{8}{9}\right)$.

All three curves also pass through the origin.



Checks

- At $\theta = 0$ the curve $r = 2(1-\sin\theta)$ is at $(2, 0)$, and $r = 2(1+\sin\theta)$ passes through the same point. There $f = 2$, $f' = -2$, $g = 2$, $g' = 2$, so $f'g' + fg = -4 + 4 = 0$, as it should be.
- At $\theta = -\frac{1}{2}\pi$, $f = 2(1+1) = 4$, as required.
- The tangents at $\left(\frac{8\sqrt{2}}{9}, \frac{4}{9}\right) \approx (1.257, 0.444)$ were confirmed perpendicular by computer, as were the tangents for many random values of a.

Common slips

- Using the right-angle condition only at one meeting point. The point of "for each positive a" is that the condition then holds at every point of C.

- Forgetting that $f(\theta)/(1-\sin\theta)$ is undefined at $\theta = \frac{1}{2}\pi$, and so being careless about where the constant is fixed.
- Drawing $r = 2(1-\sin\theta)$ opening upwards: since $r = 4$ at $\theta = -\frac{1}{2}\pi$, the curve lies mainly below the x-axis.

Beyond STEP

A family of curves that meets another family at right angles everywhere is called a pair of *orthogonal trajectories*. Here the cardioids $r = a(1+\sin\theta)$ and $r = b(1-\sin\theta)$ form such a pair. The idea appears in first-year differential equations and again in physics, where field lines cross equipotentials at right angles. Boyce and DiPrima, *Elementary Differential Equations*, covers how to find orthogonal trajectories. Needham, *Visual Complex Analysis*, shows how conformal maps produce such families.

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